



**THE CATHOLIC UNIVERSITY OF EASTERN AFRICA  
FACULTY OF SCIENCE**

**DEPARTMENT OF COMPUTER SCIENCE**

**UNIT: CMT 429**

**UNIT TITLE: INTRODUCTION TO DATA SCIENCE**

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# Phase 1 Project Report

**Domain:** Business (Bank Marketing)

**Dataset:** [Bank Marketing Dataset \(UCI Machine Learning Repository\)](#)

## Project Overview

This project focuses on using classification and regression techniques to analyze the Bank Marketing dataset from a Portuguese banking institution. The dataset captures results of direct marketing campaigns (via phone calls) where the main goal was to determine whether a client would subscribe to a term deposit. Using the R programming language, we aim to build predictive models that will support decision-making in future marketing campaigns.

## Business Understanding

### Business Problem

Banks spend significant resources on marketing campaigns, but most calls end without success. This leads to wasted time, money, and reduced customer trust. The main problem is how to predict which clients are more likely to subscribe to a term deposit, so that campaigns can be more targeted and cost-effective.

### Stakeholders

- **Bank management:** Needs insights to improve campaign efficiency.
- **Marketing teams:** Want to know which customers to prioritize.
- **Customers:** Benefit from receiving more relevant and personalized offers.

## Research Questions

1. What customer demographic and financial characteristics influence subscription decisions?
2. How do campaign-related variables affect customer response?
3. Can we build a model that accurately predicts whether a client will subscribe to a term deposit?
4. Which features are the most important for prediction?

## Justification

The need for predictive modeling in banking is crucial because it reduces unnecessary calls, improves customer experience, and increases return on investment in marketing. Data science provides tools to analyze historical data and develop models that support smarter decision-making.

## **Data Description**

- **Source:** UCI Machine Learning Repository – Bank Marketing Dataset.
- **Instances:** 45,211 records (depending on dataset version used).
- **Key Features:** Age, job, marital status, education, default, balance, housing loan, personal loan, contact type, campaign details, and previous campaign outcomes.
- **Target Variable:** y (binary: “yes” = client subscribed, “no” = client did not subscribe).
- **Data Type:** Mix of categorical, integer, and binary features.
- **Missing Values:** Minimal, mostly in categorical variables such as “pdays” and “poutcome”.

## **Objectives**

### **Main Objective**

To build a predictive classification model that determines whether a client will subscribe to a term deposit.

### **Specific Objectives**

1. Perform **data preprocessing** (cleaning, handling missing values, encoding).
2. Conduct **exploratory data analysis (EDA)** to identify trends and patterns.
3. Build and train classification models (e.g., logistic regression, decision tree, random forest).
4. Evaluate model performance using **accuracy, precision, recall, and F1-score**.
5. Perform **cross-validation** to ensure model robustness.
6. Provide **conclusions and recommendations** for banking decision-makers.

## **Exploratory Data Analysis (EDA)**

### **I. Univariate Analysis (Expected):**

- A. Distribution of age, balance, campaign contacts.
- B. Frequency of categorical variables such as job type, marital status, and education.

### **II. Bivariate Analysis (Expected):**

- A. Relationship between y (subscription) and independent features.
- B. Example: Subscription rate vs. education level, or vs. contact type.

### **III. Multivariate Analysis (Expected):**

- A. Interaction between multiple features such as job, education, and balance in predicting y.
- B. Use of correlation plots and dimensionality reduction techniques.

## **Model Overview**

- I. **Models to be tested:** Logistic Regression, Decision Tree, Random Forest, Gradient Boosting.
- II. **Evaluation Metrics:**
  - A. **Accuracy:** Percentage of correct predictions.
  - B. **Precision:** Correct positive predictions out of all predicted positives.
  - C. **Recall:** Ability to capture all true positives.
  - D. **F1-Score:** Balance between precision and recall.

## **Conclusion**

Demographic and financial factors such as age, balance, job, and education strongly influence the likelihood of subscription. Campaign-related features like number of contacts and previous campaign outcomes are critical predictors. Banks should focus marketing calls on customers with higher probability of subscription. Models can guide decision-makers to optimize resources and avoid unnecessary customer interactions.